

## SAFETY DATA SHEET

Creation Date 18-Sep-2014

Revision Date 24-Dec-2021

Revision Number 5

### 1. Identification

**Product Name** DL-1,4-Dithiothreitol

**Cat No. :** AC165680000

**CAS No** 3483-12-3  
**Synonyms** Cleland's reagent; DTT; threo-1,4-Dimercapto-2,3-butanediol

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

##### **UK entity/business name**

Fisher Scientific UK  
Bishop Meadow Road,  
Loughborough, Leicestershire LE11 5RG,  
United Kingdom  
General info; Tel: +44 (0)1509 231166

##### **EU entity/business name**

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**Emergency Telephone Number** For information **US** call: 001-800-ACROS-01 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

This chemical is considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

|                                                  |            |
|--------------------------------------------------|------------|
| Acute oral toxicity                              | Category 4 |
| Skin Corrosion/Irritation                        | Category 2 |
| Serious Eye Damage/Eye Irritation                | Category 2 |
| Specific target organ toxicity (single exposure) | Category 3 |
| Target Organs - Respiratory system.              |            |

|                  |     |
|------------------|-----|
| Combustible dust | Yes |
|------------------|-----|

**Label Elements****Signal Word**

Warning

**Hazard Statements**

May form combustible dust concentrations in air

Harmful if swallowed

Causes skin irritation

Causes serious eye irritation

May cause respiratory irritation

**Precautionary Statements****Prevention**

Wash face, hands and any exposed skin thoroughly after handling

Do not eat, drink or smoke when using this product

Wear protective gloves/protective clothing/eye protection/face protection

Avoid breathing dust/fume/gas/mist/vapors/spray

Use only outdoors or in a well-ventilated area

**Inhalation**

IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing

Call a POISON CENTER or doctor/physician if you feel unwell

**Skin**

IF ON SKIN: Wash with plenty of soap and water

If skin irritation occurs: Get medical advice/attention

Take off contaminated clothing and wash before reuse

**Eyes**

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

If eye irritation persists: Get medical advice/attention

**Ingestion**

IF SWALLOWED: Call a POISON CENTER or doctor/physician if you feel unwell

Rinse mouth

**Storage**

Store in a well-ventilated place. Keep container tightly closed

Store locked up

**Disposal**

Dispose of contents/container to an approved waste disposal plant

**Hazards not otherwise classified (HNOC)****Other hazards**

Stench.

**3. Composition/Information on Ingredients**

| Component                                 | CAS No    | Weight % |
|-------------------------------------------|-----------|----------|
| 2,3-Butanediol, 1,4-dimercapto-, (R*,R*)- | 3483-12-3 | >95      |

#### 4. First-aid measures

|                                            |                                                                                                                                                                                                                                                                                                              |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eye Contact</b>                         | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.                                                                                                                                                                                              |
| <b>Skin Contact</b>                        | Wash off immediately with plenty of water for at least 15 minutes. Get medical attention if symptoms occur.                                                                                                                                                                                                  |
| <b>Inhalation</b>                          | Remove to fresh air. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Get medical attention. If not breathing, give artificial respiration. |
| <b>Ingestion</b>                           | Do NOT induce vomiting. Call a physician or poison control center immediately.                                                                                                                                                                                                                               |
| <b>Most important symptoms and effects</b> | No information available.                                                                                                                                                                                                                                                                                    |
| <b>Notes to Physician</b>                  | Treat symptomatically                                                                                                                                                                                                                                                                                        |

#### 5. Fire-fighting measures

|                                         |                                                                                       |
|-----------------------------------------|---------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | Water spray, carbon dioxide (CO <sub>2</sub> ), dry chemical, alcohol-resistant foam. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                                                              |
| <b>Flash Point</b>                      | > 110 °C / > 230 °F                                                                   |
| <b>Method -</b>                         | No information available                                                              |
| <b>Autoignition Temperature</b>         | Not applicable                                                                        |
| <b>Explosion Limits</b>                 |                                                                                       |
| <b>Upper</b>                            | No data available                                                                     |
| <b>Lower</b>                            | No data available                                                                     |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                              |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                              |

#### Specific Hazards Arising from the Chemical

Fine dust dispersed in air may ignite. Thermal decomposition can lead to release of irritating gases and vapors. Keep product and empty container away from heat and sources of ignition.

#### Hazardous Combustion Products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Sulfur oxides. Sulfides.

#### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

#### NFPA

**Health**  
2

**Flammability**  
1

**Instability**  
1

**Physical hazards**  
N/A

#### 6. Accidental release measures

|                                             |                                                                                                                                              |
|---------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>                 | Use personal protective equipment as required. Ensure adequate ventilation. Avoid dust formation. Avoid contact with skin, eyes or clothing. |
| <b>Environmental Precautions</b>            | Avoid release to the environment. See Section 12 for additional Ecological Information.                                                      |
| <b>Methods for Containment and Clean Up</b> | Sweep up and shovel into suitable containers for disposal. Avoid dust formation.                                                             |

## 7. Handling and storage

|                 |                                                                                                                                                                                                                                                                |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid dust formation. Avoid contact with skin, eyes or clothing. Do not breathe (dust, vapor, mist, gas). Do not ingest. If swallowed then seek immediate medical assistance. |
| <b>Storage.</b> | Keep in a dry place. Keep container tightly closed. Keep under nitrogen. Keep refrigerated. Incompatible Materials. Bases. Metals. Reducing Agent.                                                                                                             |

## 8. Exposure controls / personal protection

|                                      |                                                                                                                                                                             |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Exposure Guidelines</b>           | This product does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies.                               |
| <b>Engineering Measures</b>          | Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location.                           |
| <b>Personal Protective Equipment</b> |                                                                                                                                                                             |
| <b>Eye/face Protection</b>           | Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166. |
| <b>Skin and body protection</b>      | Wear appropriate protective gloves and clothing to prevent skin exposure.                                                                                                   |
| <b>Respiratory Protection</b>        | No protective equipment is needed under normal use conditions.                                                                                                              |
| <b>Hygiene Measures</b>              | Handle in accordance with good industrial hygiene and safety practice.                                                                                                      |

## 9. Physical and chemical properties

|                                               |                                                              |
|-----------------------------------------------|--------------------------------------------------------------|
| <b>Physical State</b>                         | Powder Solid                                                 |
| <b>Appearance</b>                             | White                                                        |
| <b>Odor</b>                                   | Stench                                                       |
| <b>Odor Threshold</b>                         | No information available                                     |
| <b>pH</b>                                     | 5.1 (1 %)                                                    |
| <b>Melting Point/Range</b>                    | 39 - 44 °C / 102.2 - 111.2 °F                                |
| <b>Boiling Point/Range</b>                    | 125 - 130 °C / 257 - 266 °F @ 2 mmHg                         |
| <b>Flash Point</b>                            | > 110 °C / > 230 °F                                          |
| <b>Evaporation Rate</b>                       | Not applicable                                               |
| <b>Flammability (solid,gas)</b>               | No information available                                     |
| <b>Flammability or explosive limits</b>       |                                                              |
| Upper                                         | No data available                                            |
| Lower                                         | No data available                                            |
| <b>Vapor Pressure</b>                         | No information available                                     |
| <b>Vapor Density</b>                          | Not applicable                                               |
| <b>Specific Gravity</b>                       | No information available                                     |
| <b>Solubility</b>                             | completely soluble                                           |
| <b>Partition coefficient; n-octanol/water</b> | No data available                                            |
| <b>Autoignition Temperature</b>               | Not applicable                                               |
| <b>Decomposition Temperature</b>              | No information available                                     |
| <b>Viscosity</b>                              | Not applicable                                               |
| <b>Molecular Formula</b>                      | C <sub>4</sub> H <sub>10</sub> O <sub>2</sub> S <sub>2</sub> |
| <b>Molecular Weight</b>                       | 154.25                                                       |

## 10. Stability and reactivity

|                                         |                                                                                                  |
|-----------------------------------------|--------------------------------------------------------------------------------------------------|
| <b>Reactive Hazard</b>                  | None known, based on information available                                                       |
| <b>Stability</b>                        | Moisture sensitive. Air sensitive.                                                               |
| <b>Conditions to Avoid</b>              | Avoid dust formation. Incompatible products. Excess heat. Exposure to air. Exposure to moisture. |
| <b>Incompatible Materials</b>           | Bases, Metals, Reducing Agent                                                                    |
| <b>Hazardous Decomposition Products</b> | Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> ), Sulfur oxides, Sulfides                 |
| <b>Hazardous Polymerization</b>         | No information available.                                                                        |
| <b>Hazardous Reactions</b>              | None under normal processing.                                                                    |

## 11. Toxicological information

### Acute Toxicity

#### Product Information Component Information

| Component                                 | LD50 Oral         | LD50 Dermal | LC50 Inhalation |
|-------------------------------------------|-------------------|-------------|-----------------|
| 2,3-Butanediol, 1,4-dimercapto-, (R*,R*)- | 400 mg/kg ( Rat ) | Not listed  | Not listed      |

**Toxicologically Synergistic Products** No information available

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

**Irritation** Irritating to eyes, respiratory system and skin

**Sensitization** No information available

**Carcinogenicity** The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Component                                 | CAS No    | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|-------------------------------------------|-----------|------------|------------|------------|------------|------------|
| 2,3-Butanediol, 1,4-dimercapto-, (R*,R*)- | 3483-12-3 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** Respiratory system  
**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** No information available

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

Do not empty into drains.

**Persistence and Degradability** Soluble in water Persistence is unlikely based on information available.

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Will likely be mobile in the environment due to its water solubility.

### 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

### 14. Transport information

#### DOT

**UN-No** UN3335  
**Proper Shipping Name** AVIATION REGULATED SOLID, N.O.S.  
**Hazard Class** 9

#### TDG

**UN-No** UN3335  
**Proper Shipping Name** AVIATION REGULATED SOLID, N.O.S.  
**Hazard Class** 9

#### IATA

**UN-No** UN3335  
**Proper Shipping Name** AVIATION REGULATED SOLID, N.O.S.\*  
**Hazard Class** 9

**IMDG/IMO** Not regulated

### 15. Regulatory information

#### United States of America Inventory

| Component                                 | CAS No    | TSCA | TSCA Inventory notification - Active-Inactive | TSCA - EPA Regulatory Flags |
|-------------------------------------------|-----------|------|-----------------------------------------------|-----------------------------|
| 2,3-Butanediol, 1,4-dimercapto-, (R*,R*)- | 3483-12-3 | X    | ACTIVE                                        | -                           |

#### **Legend:**

**TSCA** US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

X - Listed

'-' - Not Listed

**TSCA 12(b)** - Notices of Export Not applicable

#### **International Inventories**

Canada (DSL/NDL), Europe (EINECS/ELINCS/NLP), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), China (IECSC), Korea (KECL).

| Component                                 | CAS No    | DSL | NDL | EINECS    | PICCS | ENCS | ISHL | AICS | IECSC | KECL |
|-------------------------------------------|-----------|-----|-----|-----------|-------|------|------|------|-------|------|
| 2,3-Butanediol, 1,4-dimercapto-, (R*,R*)- | 3483-12-3 | X   | -   | 222-468-7 | X     | -    |      | X    | X     | -    |

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

#### U.S. Federal Regulations

**SARA 313** Not applicable

**SARA 311/312 Hazard Categories** See section 2 for more information

|                                                             |                                                             |
|-------------------------------------------------------------|-------------------------------------------------------------|
| <b>CWA (Clean Water Act)</b>                                | Not applicable                                              |
| <b>Clean Air Act</b>                                        | Not applicable                                              |
| <b>OSHA - Occupational Safety and Health Administration</b> | Not applicable                                              |
| <b>CERCLA</b>                                               | Not applicable                                              |
| <b>California Proposition 65</b>                            | This product does not contain any Proposition 65 chemicals. |
| <b>U.S. State Right-to-Know Regulations</b>                 | Not applicable                                              |
| <b>U.S. Department of Transportation</b>                    |                                                             |
| Reportable Quantity (RQ):                                   | N                                                           |
| DOT Marine Pollutant                                        | N                                                           |
| DOT Severe Marine Pollutant                                 | N                                                           |
| <b>U.S. Department of Homeland Security</b>                 | This product does not contain any DHS chemicals.            |

**Other International Regulations**

**Mexico - Grade** No information available

**Authorisation/Restrictions according to EU REACH****Safety, health and environmental regulations/legislation specific for the substance or mixture**

| Component                                 | CAS No    | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|-------------------------------------------|-----------|----------------|------------------------------|---------------------------|--------------------------------------------|
| 2,3-Butanediol, 1,4-dimercapto-, (R*,R*)- | 3483-12-3 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |

| Component                                 | CAS No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|-------------------------------------------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| 2,3-Butanediol, 1,4-dimercapto-, (R*,R*)- | 3483-12-3 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

## 16. Other information

|                         |                                                                                                                                                                                                                                         |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Prepared By</b>      | Regulatory Affairs<br>Thermo Fisher Scientific<br>Email: EMSDS.RA@thermofisher.com                                                                                                                                                      |
| <b>Creation Date</b>    | 18-Sep-2014                                                                                                                                                                                                                             |
| <b>Revision Date</b>    | 24-Dec-2021                                                                                                                                                                                                                             |
| <b>Print Date</b>       | 24-Dec-2021                                                                                                                                                                                                                             |
| <b>Revision Summary</b> | This document has been updated to comply with the US OSHA HazCom 2012 Standard replacing the current legislation under 29 CFR 1910.1200 to align with the Globally Harmonized System of Classification and Labeling of Chemicals (GHS). |

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**